

Planet Search Roadmap

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This document outlines the process of running QLP's planet search pipeline on the pdo compute system. The process starts when QLP light curves for all orbits in a sector have been produced, processed, and ingested into the light curve database. After completing these steps, operators should head to the roadmap on delivering to TEV.

Summary: Just the commands

This section lists the command templates used in order for experienced operators. If this is your first time operating, or if you run into problems, find more details in subsequent sections.

```
cd /pdo/qlp-data
./new-qlp-sector.sh $SECTOR
for cam in {1..4}; do for ccd in {1..4}; do ls /pdo/qlp-data/orbit-$ORBIT/ffi/cam${cam}/ccd${ccd}/LC | awk '
  BEGIN {
    CUDA_DEVICE=$NUM
    qlp lctools bls -c bls.cfg -i cam${CAM}ccd${CCD}.ls --cam $CAM --ccd $CCD --source db --s
    qlp estools tic-filter -c cutoffs.cfg -o filter_cam${CAM}.ls -s $SECTOR --autolist $CAM -n 72
    qlp estools astronnet --triage -i filter_cam${CAM}.ls -s $SECTOR -n 72 -o astronnet_scores_cam${CAM}.csv
    qlp estools astronnet-filter -i astronnet_scores_cam${CAM}.csv -o astronnet_cam${CAM}.ls --score-col disp_E
    qlp estools difference-image -i astronnet_cam${CAM}.ls -s $SECTOR -n 72
    qlp estools diffimage-centroid-filter -i astronnet_cam${CAM}.ls -s $SECTOR -o centroid_cam${CAM}.ls
    qlp estools astronnet --vetting -i centroid_cam${CAM}.ls -s $SECTOR -n 72 -o astronnet_vetting_scores_cam${CAM}.csv
    qlp estools astronnet-filter -i astronnet_vetting_scores_cam${CAM}.csv -o astronnet_vetting_cam${CAM}.ls --s
    qlp estools report -i astronnet_cam${CAM}.ls -s $SECTOR -n 72 -p 0 3 5 7 --nok2
  }
  feh -f pnglist.txt --action1 "echo %n >> accept.ls" --action2 "echo %n >> reject.ls" [optional resize: --
```

Sector setup

A script in `/pdo/qlp-data` sets up a new sector directory with proper permissions and configuration.

```
cd /pdo/qlp-data
./new-qlp-sector.sh $SECTOR
```

Tip

Running the script with no arguments shows the proper usage, if you ever forget.

BLS Search

QLP's BLS search is designed to be run on a GPU machine (pdogpu*). Machines with RTX6000 GPUs (pdogpu 1,2,3) have demonstrated the best performance historically.

To run the BLS search, prepare a list of TIC IDs of stars that have light curves in the current sector. Then run QLP's BLS command, generating result files in the `MP/` ("multi planet") directory for each camera/CCD.

```
for cam in {1..4}; do for ccd in {1..4}; do ls /pdo/qlp-data/orbit-$ORBIT/ffi/cam${cam}/ccd${ccd}/LC | awk '
  BEGIN {
    CUDA_DEVICE=$NUM
    qlp lctools bls -c bls.cfg -i cam${CAM}ccd${CCD}.ls --cam $CAM --ccd $CCD --source uv --s
```

Notes:

- Options for the BLS config can be found on the [BLS Options](#) wiki page.
- `CUDA_DEVICE=$NUM` specifies which physical GPU should be used for processing. If you are running multiple BLS commands on the same machine (e.g., in multiple tmux panes), they should use different GPUs so they don't oversubscribe the device.
- `--read-fraction 0.1` tells QLP to use ~10% of the processors for pulling down light curves form the database (as opposed to working on BLS searches). This is a balance - if the queue length is always 0, then worker processes are sitting idle, but we also don't want to have too few worker processes.
- Based on testing, we can have ~400 processes running between all simultaneous calls to `qlp lctools bls`. This translates to 4 CCDs simultaneously with ~100 processes each, or all 16 CCDs simultaneously with ~30 processes each. We are limited by database connections, which are capped at 500 for both the light curve database (used only by the "reader" processes) and the data product tracker (used by all processes).

Planet signal triage

QLP's triage process starts with the list of all possible candidate signals and then eliminates non-planet-like candidates in a series of tests. The file naming convention for these lists is as follows:

- `filter_cam${CAM}.ls` contains the list of candidates that passed the BLS/TIC filter (`qlp estools tic-filter`).
- `astronnet_cam${CAM}.ls` contains the list of candidates that pass the astronnet-triage threshold (scores are saved in `astronnet_scores_cam${CAM}.csv`).
- `centroid_cam${CAM}.ls` contains the list of candidates that pass the difference image centroid offset cutoff.
- `astronnet_vetting_cam${CAM}.ls` contains the list of candidates that pass the astronnet-vetting threshold (scores are save in `astronnet_vetting_scores_cam${CAM}.csv`).
- `pcs_cam${CAM}.ls` contains the candidates that pass manual triage.

The following commands are used to carry out each of the steps. Generally there are pairs of commands where the first generates some data (astronnet scores, difference images) and the second filters a signal list based on the generated data. Note that the BLS command generates the data for the "TIC filter", which also includes a filter on BLS signal strength.

Configuration options are commented in the `cutoffs.cfg` file.

```
qlp estools tic-filter -c cutoffs.cfg -o filter_cam${CAM}.ls -s $SECTOR --autolist $CAM -n 72
qlp estools astronnet --triage -i filter_cam${CAM}.ls -s $SECTOR -n 72 -o astronnet_scores_cam${CAM}.csv
qlp estools astronnet-filter -i astronnet_scores_cam${CAM}.csv -o astronnet_cam${CAM}.ls --score-col disp_E
qlp estools difference-image -i astronnet_cam${CAM}.ls -s $SECTOR -n 72
qlp estools diffimage-centroid-filter -i astronnet_cam${CAM}.ls -s $SECTOR -o centroid_cam${CAM}.ls
qlp estools astronnet --vetting -i centroid_cam${CAM}.ls -s $SECTOR -n 72 -o astronnet_vetting_scores_cam${CAM}.csv
qlp estools astronnet-filter -i astronnet_vetting_scores_cam${CAM}.csv -o astronnet_vetting_cam${CAM}.ls --s
```

Generate visual reports

The final step in the process is to generate visual reports for human triage. Many reports are cheap to make, but some are more expensive and not worth making for an initial triage. The following reports are available:

- Page 0: summary
- Page 2: BLS spectrum
- Page 3: depth-aperture correlation
- Page 5: difference image
- Page 6: MCMC light curve fit
- Page 7: full detrended light curve, automatically generated with page 7

For human triage, it is recommended to produce pages 0, 3, 5, and 7. While human triage is being run parallel to astronnet-vetting, reports should be produced for the astronnet-triage lists (`astronnet_cam${CAM}.ls`); in the future, reports only need to be produced for the astronnet-vetting lists (`astronnet_vetting_cam${CAM}.ls`).

Note

Page "7" is now only made per *star* as it will contain all the relevant TCEs within the report. This is to reduce the total amount of PDF pages emitted by very long lightcurve baselines,

```
qlp estools report -i astronnet_cam${CAM}.ls -s $SECTOR -n 72 -p 0 3 5 7 --nok2
```

At this point, reports can be manually triaged. The following command will bring up the reports listed in a text file (`pnglist.txt`) and allow to press `1` to append the report file path to `accept.ls` and `2` to append it to `reject.ls`. The list of accepted candidates can then be inferred from the file names in the accept list.

```
feh -f pnglist.txt --action1 "echo %n >> accept.ls" --action2 "echo %n >> reject.ls" [optional resize: --
```

Finally, compile a list of final planet candidates for all cameras in `pcs.ls`.

Deliver to TEV

At this point, the QLP work is complete and the list of planet candidates is ready to be delivered to TEV. Head to the [TEV Delivery](#) wiki page for more details on that process.

For now first, ask Glen, Katharine or Willie to create a collection on TEV with the name `qlp-s${SECTOR}-ffi` (e.g., `qlp-s94-ffi`).

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